

Cross-Type Identity-Document Fraud Detection: Fourier Domain Adaptation and Architecture-Diverse Ensembling

The FREUID Challenge 2026: Technical Report (Team MobilityAI)

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Abstract

The FREUID Challenge 2026 requires a scalar fraud score in $[0, 1]$ for each identity-document image, ranked by the FREUID Score (AuDET + APCER@1%BPCER; lower is better). The private split, which decides the ranking, is born-digital and contains identity-document types that are not present in the training data. The central difficulty is therefore cross-type generalization: a detector must judge a document type it has never seen without flagging a genuine one simply because it looks unfamiliar. Our submitted model is the probability average of two full fine-tuned RGB detectors, a ViT-L/14 DINOv2 transformer and a ConvNeXt-V2-L convolutional network, both trained at 896 pixels on the FREUID training split only, with no external data. Two design choices target unseen types. First, Fourier Domain Adaptation randomizes the low-frequency global appearance during training, which decouples the fraud signal from document-type look. Second, we combine a transformer and a convolutional backbone, which rely on different cues and fail on different document types, so their average holds up where either alone would slip. A tail-margin loss focuses training on the low-BPCER tail that the metric scores. In leave-one-document-type-out cross-validation on the two hardest folds, the two backbones are complementary (prediction correlation about 0.78 to 0.85), and their average lowers the two-fold mean FREUID Score from about 0.0084 for a single backbone to 0.0032, reaching 0.0002 on the EGYPT fold. Inference is a plain probability average in a pinned container that runs offline.

Competition: The FREUID Challenge 2026 (IJCAI-ECAI 2026), Kaggle [the-freuid-challenge-2026-ijcai-ecai](https://kaggle.com/freuid-challenge-2026-ijcai-ecai) [6]

Kaggle team: MobilityAI

Code: <https://github.com/hoppery/freuid-challenge-2026>

1 Introduction

Identity-document fraud detection asks whether a document image is genuine or fraudulent. The FREUID Challenge [6] frames this as a single scalar task: the model outputs $P(\text{fraudulent}) \in [0, 1]$ for each image, scored by the FREUID Score (AuDET + APCER@1%BPCER; lower is better). The APCER@1%BPCER term measures the attack-acceptance rate at a strict 1% genuine-rejection operating point, so the metric rewards a model that keeps a very low false-positive tail on genuine documents.

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The property that shapes our design is the gap between training and the private test in document type. The training data covers five document types (EGYPT, GUINEA, BENIN, MOZAMBIQUE, and MAURITIUS driving licence and identity cards) and is born-digital. The private split, which decides the ranking, is also born-digital but includes document types that are not in the training data. A detector that memorizes the exact appearance of the five training types can look strong in validation and still fail on the private split, because a genuine document of an unseen type is unfamiliar and is easy to mistake for a manipulation. Cross-type generalization is therefore the goal, not fitting the seen types more tightly.

We address this with one model and two design choices. Fourier Domain Adaptation [8] randomizes the low-frequency global appearance of training images, which pushes the network to decide on structure and local evidence rather than on the overall color and layout of a given type. Architecture diversity pairs a self-supervised vision transformer (ViT-L/14 DINOv2 [5, 1]) with a convolutional backbone (ConvNeXt-V2-L). The two families carry different inductive biases, make different mistakes across document types, and are only weakly correlated, so a plain probability average is more stable on unseen types than either model alone [2].

Contributions.

1. We build a cross-type fraud detector from FREUID training data only, with no external data, using Fourier Domain Adaptation to decouple the fraud signal from document-type appearance (Section 2).
2. We show that a transformer and a convolutional backbone are complementary across document types: each is strong on the fold where the other is weak, and their predictions are decorrelated (Section 5).
3. We show, by leave-one-document-type-out cross-validation on the two hardest folds, that a plain probability average of the two backbones generalizes better across types than either single model (Section 5, Fig. 2).

Related work. Identity-document forgery detection and presentation-attack detection are active benchmark areas. For synthetic and generative content, adapter-based detectors on frozen vision-language features generalize across generators [4]. Robustness to appearance shift has been approached by randomizing feature statistics, for example MixStyle [9], and by randomizing low-frequency image content, as in Fourier Domain Adaptation [8], which we adopt. We use self-supervised transformer features (DINOv2 [5], ViT [1]) together with a modern convolutional backbone (ConvNeXt-V2-L), both from `timm` [7]. Because the metric scores a strict operating point, we train with a loss that focuses on the hard low-BPCER tail, related in spirit to focal loss [3]. Our ensembling builds on deep ensembles [2]; we add evidence that cross-architecture diversity, rather than same-family seed diversity, is what helps on unseen document types.

2 Method

2.1 Overview and problem setting

Let x be an image and $y \in \{0, 1\}$ its fraud label. Each detector is an RGB backbone with a single-logit head. It produces a fraud logit $\ell(x)$ and a probability $\sigma(\ell(x)) \in [0, 1]$, where σ is the sigmoid. The submitted score is the arithmetic mean of the two detectors' probabilities. Training data is born-digital and covers the five seen document types; the private evaluation adds document types that are not in training. Because the private data is not available at submission time, we do not fit to it; instead we select the recipe by leave-one-document-type-out cross-validation on the training types (Section 3). Every detector uses the same feed-forward pipeline: decode, resize to 896 pixels with ImageNet normalization, backbone, single-logit head, and sigmoid (Fig. 1).

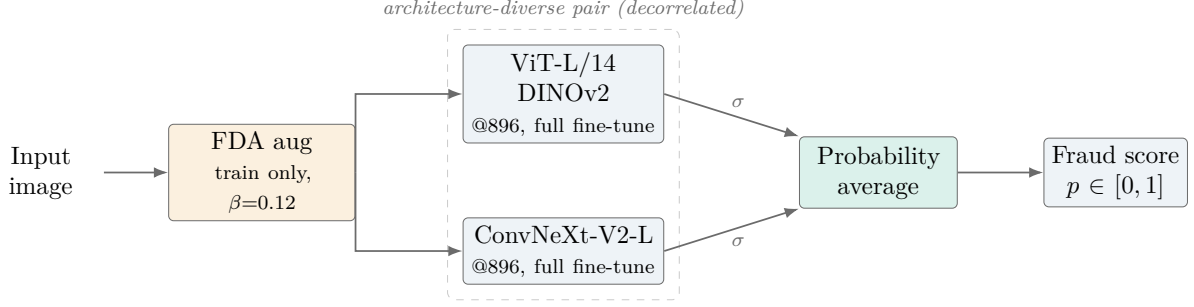


Figure 1: The submitted unseen-type detector. Each training image passes Fourier Domain Adaptation, which randomizes its low-frequency global appearance. Two full fine-tuned backbones, a transformer and a convolutional network, each output a fraud probability; the submission score is their arithmetic mean. The two families are decorrelated, so the average is more stable on unseen document types than either alone.

2.2 The two detectors

The submitted model averages two detectors that share the same recipe and differ only in backbone family.

- **ViT-L/14 DINOv2:** backbone `vit_large_patch14_reg4_dinov2.lvd142m`, a self-supervised vision transformer, at 896 pixels (`checkpoints/exp_fdab12_all/epoch2.pt`).
- **ConvNeXt-V2-L:** backbone `convnextv2_large.fcmae_ft_in22k_in1k`, a convolutional network, at 896 pixels (`checkpoints/exp_cnx FDA_all/epoch2.pt`).

Both are `model_type=rgb`: the backbone feeds a single linear logit that a sigmoid maps to $P(\text{fraud})$. There is no forensic residual branch and no auxiliary head. Both backbones are fully fine tuned (the training config sets `freeze_backbone=false`). We choose two families on purpose. A self-supervised transformer and a convolutional network use different receptive fields and different global context, so they tend to fail on different document types. That difference is what makes their average robust on types neither has seen (Section 5).

2.3 Fourier Domain Adaptation

The main lever for cross-type generalization is Fourier Domain Adaptation (FDA) [8], applied as a training augmentation. For a training image, we replace the low-frequency band of its amplitude spectrum with that of another training image (the donor), keeping the original phase, and invert back to an image. The band width is controlled by $\beta = 0.12$. Amplitude at low frequencies carries the global color and layout of a document type, while phase and high-frequency amplitude carry structure and fine detail. Swapping the low-frequency amplitude therefore randomizes the overall look of the document while preserving the structure and local evidence that separate genuine from fraudulent. The network is pushed to decide on cues that transfer across types, so a genuine document of an unseen type is less likely to be flagged only because its global appearance is unfamiliar.

2.4 Tail-margin operating-point loss

The FREUID Score includes APCER@1%BPCER, the attack-acceptance rate at a strict 1% genuine-rejection operating point. Plain binary cross-entropy optimizes average separation, not this tail. We therefore train with a tail-margin loss that adds a margin penalty on the hardest genuine examples near the operating threshold, so the model pushes the high-scoring genuine tail below the threshold that the metric reads. In probability space (`tail_space=prob`), the loss uses a tail quantile of 0.99 to locate the threshold region, a margin of 1.0, a weight $\lambda = 1.0$, an offset $\mu = 0.2$, and a 300-step warmup before the tail term is enabled (Appendix A). The full

objective is the fraud binary cross-entropy plus this tail-margin term. This loss does not change the architecture and adds no inference cost.

2.5 Architecture-diverse ensembling

A single backbone is strong on some document types and weak on others (Section 5). We reduce this variance across types by averaging the two decorrelated backbones. The averaging is a plain arithmetic mean of the two probabilities, with no weighting and no learned combiner. We do not add same-family seed copies, because same-family models are highly correlated and their average barely moves the score. The diversity that helps here is across architecture family, where the two members disagree on different document types and the average cancels each member’s type-specific errors.

2.6 Score calibration

Each detector outputs a fraud logit that a sigmoid maps to a per-model probability. The submission score is the arithmetic mean of the two probabilities. We apply no temperature and no learned calibration, and no post-processing differs between training and test time. Images that fail to decode are assigned 0.5, so every input gives exactly one finite score in $[0, 1]$.

3 Data

We use the official FREUID `train/` images and `train.csv` metadata (Kaggle competition `the-freuid-challenge-2026-ijcai-ecai`), assembled into a single manifest `manifests/freuid.parquet`. The submitted model trains on this FREUID training split only. It uses **no external data**: no captured-document corpora, no extra fraud sources, and no test data at any point. The pretrained backbone weights are used under their upstream licenses.

Validation by leave-one-document-type-out. The private split contains document types that are not in training, so a random held-out split of the training types would over-report cross-type performance. To estimate cross-type generalization with the data we have, we use leave-one-document-type-out (LODO) cross-validation: we hold out one training document type entirely, train on the other four, and score on the held-out type. This simulates the unseen-type condition, because the held-out type is absent from training. We ran LODO across the training types and report the two hardest folds, EGYPT and BENIN, in Section 5. The other held-out folds (GUINEA, MOZAMBIQUE, and MAURITIUS) were easier and are not the bottleneck; we do not report per-fold numbers for them here. For run logging we also hold out a random 8% of the manifest during each training run (`val_fraction=0.08`). LODO is an estimate on training types held out one at a time, not a sample of the private distribution, so it indicates the direction of cross-type generalization rather than a private score.

4 Inference

The container entrypoint indexes a flat directory of images. It builds each checkpoint with `pretrained=False`, loads the embedded weights, runs a forward pass, and applies a sigmoid. Preprocessing is a plain resize to 896 pixels with ImageNet normalization; there is no tiling, no multi-crop, and no test-time augmentation. For each image it averages the two per-model probabilities and writes `id,label`, where `label` is the average fraud probability in $[0, 1]$. The device is CUDA if available, else CPU. Images that fail to decode are assigned 0.5. The exact command is in Section 6.

Table 1: Leave-one-document-type-out FREUID Score $\downarrow$ (lower is better) on the two hardest folds. Each single backbone is strong on one fold and weak on the other; the probability-average ensemble is low on both and gives the best two-fold mean. The ensemble BENIN entry (0.0062) is the value implied by the measured EGYPT score and the measured two-fold mean.

Model	EGYPT $\downarrow$	BENIN $\downarrow$	Two-fold mean $\downarrow$
ViT-L/14 DINOv2 (FDA)	0.0020	0.0140	0.0080
ConvNeXt-V2-L (FDA)	0.0150	0.0023	0.0086
Ensemble (prob. average)	0.0002	0.0062	0.0032

5 Results

We report leave-one-document-type-out (LODO) FREUID Scores (lower is better) on the two hardest folds. The two folds are hard for opposite reasons. EGYPT is appearance out-of-distribution: it has a distinct script and color palette, so a genuine EGYPT document is easy to flag as unfamiliar. BENIN carries subtle chroma-dependent attacks, where the separation between genuine and fraud lives in the color channels. No single uniform treatment fixes both at once, which is why we look for two backbones that fail differently.

5.1 The two backbones are complementary

Table 1 and Fig. 2 show the pattern. The ViT-L/14 DINOv2 detector is strong on EGYPT (0.0020) but weak on BENIN (0.0140). The ConvNeXt-V2-L detector is the reverse: strong on BENIN (0.0023) and weak on EGYPT (0.0150). Each model owns the fold that the other struggles with. Their predictions are decorrelated, with a correlation between about 0.78 and 0.85 across the folds, which is low enough for averaging to help. This is the cross-type version of the diversity effect: because the two families fail on different document types, neither alone is safe for a type it has not seen.

5.2 The ensemble generalizes across types

The probability average is low on both folds. On EGYPT it reaches 0.0002, below both single models, and on BENIN it stays low at 0.0062. The two-fold mean falls from about 0.0084, the mean of the two single backbones, to 0.0032 for the ensemble. The point is not that the average wins every fold, but that it does not collapse on either. A single backbone would be a risky choice for the private split, where the document type is unknown in advance, because it could land on the fold it is weak on. Averaging the two decorrelated backbones removes that risk in our LODO estimate, which is the behavior we want on unseen types. We do not claim any leaderboard rank or any private-test score; we have none, and LODO on training types is an estimate of direction, not a private measurement.

5.3 Design choices we kept and dropped

Two choices carried the cross-type result. Fourier Domain Adaptation was kept because randomizing low-frequency appearance is what lets an unseen-type genuine document avoid being flagged for looking unfamiliar. Cross-architecture averaging was kept because it is the diversity that actually moves the score: the two families disagree on different document types. We dropped same-family seed ensembling, because same-family members are highly correlated and their average barely changes the result, unlike the cross-architecture pair reported above. We also kept the model to FREUID training data only, since external data did not target the born-digital, unseen-type condition that the private split presents.

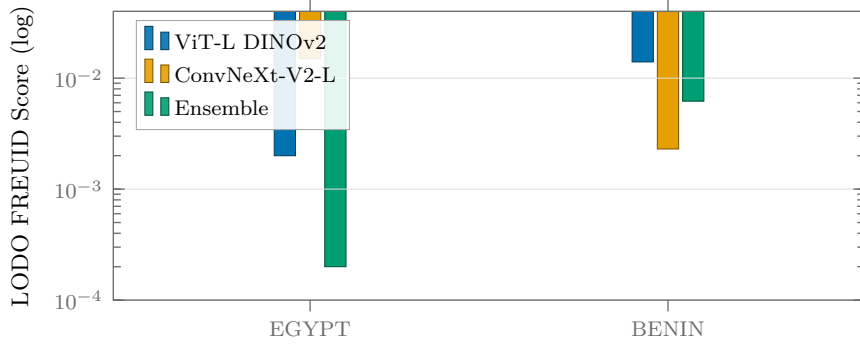


Figure 2: Leave-one-document-type-out FREUID Score on the two hardest folds (log scale, lower is better). The transformer is strong on EGYPT and weak on BENIN; the convolutional backbone is the reverse; their average is low on both. The BENIN ensemble bar (0.0062) is the value implied by the measured EGYPT score and two-fold mean.

6 Reproducibility

The reproducibility container ships the submitted two-model average. The two checkpoints are fetched from a GitHub Release at build time, verified by SHA-256, and baked into the image, and both models build with `pretrained=False`. The container therefore runs with `-network none` (no runtime downloads) and writes only to `/submissions`.

- **Environment (pinned):** `torch==2.12.0` (CUDA 13), `timm==1.0.27`, `albumentations==2.0.8`, `opencv-python-headless==4.13.0.92`, `numpy==2.4.6`, `pandas==3.0.3`, `scipy==1.17.1`, `scikit-learn==1.9.0`, `pillow==12.2.0`, `pyyaml==6.0.3`, `pyarrow==24.0.0`; Python ≥ 3.11 .
- **Container:** base image `nvidia/cuda:13.0.1-cudnn-runtime-ubuntu24.04`. Weights are fetched from the GitHub Release, SHA-256 verified, and baked in, so the image runs offline with `-network none`. Input is a flat `/data` directory of images; output is `/submissions/submission.csv` with columns `id,label`.
- **Hardware and runtime:** training on 2× NVIDIA RTX PRO 6000 (Blackwell, 96 GB); inference uses a single GPU. The base image targets CUDA 13, which needs a host NVIDIA driver $\geq 580.65.06$; on a host with an older driver the container falls back to CPU, so adjust the base tag and the `torch` index to match (the checkpoint weights are torch-version portable).
- **Randomness:** global seed 42. CUDA, cuDNN, and mixed precision add minor non-determinism.
- **Repository:** <https://github.com/hopper/freuid-challenge-2026> (MIT license). The frozen commit SHA is stated in our reproducibility reply.

Build and run (organizer contract):

```
docker build -t freuid-repro:local .
docker run --gpus all --rm --network none \
  -v /path/to/flat/test/images:/data:ro \
  -v "$(pwd)/out":/submissions \
  freuid-repro:local # -> out/submission.csv (id,label)
```

7 Team and contributions

Both authors contributed to the method, the experiments, and the manuscript. Su-Yong An (corresponding author) led model development, training, and evaluation. Lae-Kyoung Lee contributed to the methodology, the analysis, and the manuscript.

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References

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A Hyperparameters

Both detectors share one training recipe and differ only in the backbone. Each is a 6-epoch run, and we select epoch 2 of each. FDA uses $\beta = 0.12$. The tail-margin loss uses tail quantile 0.99, margin 1.0, weight $\lambda = 1.0$, offset $\mu = 0.2$, warmup 300 steps, computed in probability space.

Table 2: Per-model hyperparameters of the submitted two-model average. Shared: AdamW, learning rate 5×10^{-5} , weight decay 0.05, cosine schedule with 500 warmup steps, 6 epochs (epoch 2 selected), batch size 16 with gradient accumulation 2 (effective batch 32), bf16 mixed precision, gradient checkpointing, gradient clip 1.0, EMA with decay 0.999, and seed 42.

Hyperparameter	ViT-L/14 DINOv2	ConvNeXt-V2-L
Backbone	<code>vit_large_patch14_reg4_dinov2.lvd142m</code>	<code>convnextv2_large.fcmae_ft_in22k_in1k</code>
Model type	rgb (single logit)	rgb (single logit)
Image size	896	896
Learning rate	5×10^{-5}	5×10^{-5}
Weight decay	0.05	0.05
Batch size (eff.)	16 ($\times 2 = 32$)	16 ($\times 2 = 32$)
Epochs (used)	6 (2)	6 (2)
Backbone training	full fine-tune	full fine-tune
Loss	tail-margin	tail-margin
FDA β	0.12	0.12
Seed	42	42

B Additional notes

See Table 1 and Fig. 2 for the leave-one-document-type-out results on the two hardest folds, and Section 5.3 for the design choices we kept and dropped. Training and inference configuration files (`configs/exp_fdab12_all.yaml` and `configs/exp_cnx FDA_all.yaml`) and further run notes are in the repository.